EDA262 · DATA ENGINEERING · PART 2 (AV2)

Of the vulnerabilities already exploited, which are still likely to be exploited again?

The Part 1 lake gained real volume (EPSS), three layers, partitioned Parquet, idempotent and incremental loads, and per-layer access in Lake Formation — and answers for US$ 0.00004768.

Group g07 · CESAR School · 2026.2

Bruno Ribeiro · Emanuel Eduardo · Igor Alves · Leonardo Amaral · Victor Guilherme · Vinícius Petribu

Why EPSS: Part 1 sat below the billing floor

- Athena bills at least **10 MB** per query. All of KEV is **1.5 MB**: Parquet and partitions on it would save **US$ 0.00**.
- **EPSS** (FIRST): each CVE's probability of exploitation in the next 30 days, published **every day** for **~384k** CVEs.
- Joining them adds **information**, not just volume: among exploited CVEs, it separates the ones still dangerous now.

> **Our question:** which vendors concentrate last year's exploited CVEs — and how likely is each vendor's exploited portfolio to be exploited?

80.9 MB

30 days of EPSS (gzip) — above the floor

11.5 M

rows in the 30-day window

0.49 vs 0.0065

median EPSS: KEV CVEs vs all (76×)

Architecture: three layers, all in code

1

Collect

KEV and EPSS, by backfill or incremental load.

ingest_kev.sh · ingest_epss.sh

›

2

Raw

The bytes exactly as published.

gzip CSV + JSON · partition projection

›

3

Trusted

Validated and typed, grain guaranteed.

Parquet ZSTD · daily partitions

›

4

Refined

The ready answer: vendor × day.

Parquet ZSTD · vendor × day

> **Terraform** in 2 modules (data-lake, governance) · **Lake Formation** per layer · transformations as **SQL in Athena** · workspace **av2** · **73 resources** in one apply from scratch.

Partitioning and idempotency, measured on AWS

DAILY PARTITION (SNAPSHOT_DATE)

29×

fewer bytes for a 1-day query: **779,638 B** against **22,606,275 B** for 30 days.

- Grain **(cve_id, day)**: **0** duplicates in 11,519,635 rows.
- Point-in-time: **37** CVEs counted on the right day.

IDEMPOTENT AND INCREMENTAL LOADS

391,915

rows before and after re-loading 2026-10-20. Append-only would give **783,830**.

- 30-day backfill in a single INSERT: read the raw files **exactly once** (80.9 MB).
- Incremental from each layer's high-water mark.

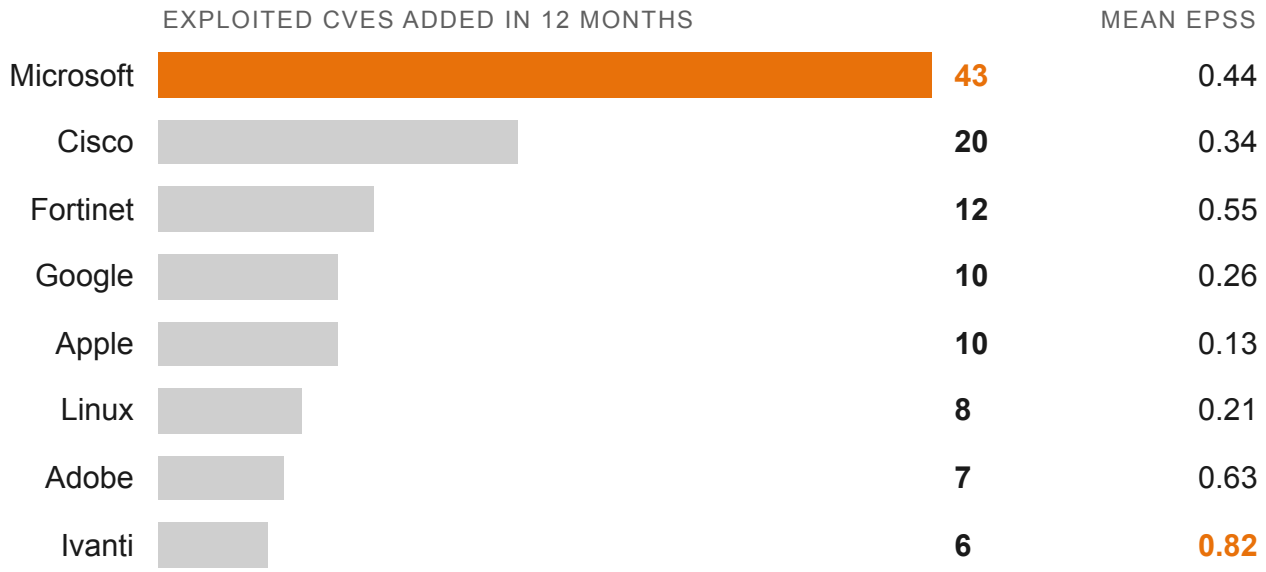
Who reads each layer: decided by Lake Formation

PERSONA	RAW	TRUSTED	REFINED
data-engineer	**reads**	**reads**	**reads**
data-analyst	denied	**reads**	**reads**
business-user	denied	denied	**reads**

Live test: every persona assumed, every layer queried. **9 of 9** match the declared matrix. A real denial: *“Insufficient Lake Formation permission(s): Required Describe on eda262_g07_raw”*.

- **Identical** IAM for all 3 personas: a denial can only come from Lake Formation.
- **0** S3 permissions on the lake buckets.
- **IAMAllowedPrincipals default removed**: otherwise the grants would be decorative.
- One WorkGroup per persona: results and history isolated.

The answer: volume is still not risk



Ranked by exploited CVEs added to KEV in the 12 months to 2026-10-20. Mean EPSS across all of each vendor's exploited CVEs, on the last day of the 30-day window.

CHANCE OF EXPLOITATION

0.82 vs 0.13

Ivanti (6 new) · Apple (10 new)

Ivanti is only 8th by volume, but its exploited portfolio is the most likely to be exploited in the ranking — 6× Apple's.

Cost target declared and met

US$ 0.00004768

per query: **162,943 B** read out of at most 10,485,760 — **64×** under target

THE SAME QUESTION, PER LAYER (BYTES READ)



The three answers came out **identical, byte for byte**.

- Refined reads **505×** less than raw; cost drops **7.9×**, because the 10 MB floor caps it.
- Athena's Parquet came out **13% larger** than the CSV and still read **1.45×** less: cost is what you read.
- Measured, not estimated: the local estimate was wrong, and we fixed it.

Demo: a real run on AWS

1 apply from scratch in workspace av2

```
Apply complete! Resources: 73 added, 0 changed, 0 destroyed.
  OK Workspace          : av2
  OK Pipeline WorkGroup : eda262-g07-wg
  OK persona data-analyst:
     eda262-g07-wg-data-analyst reads trusted, refined
```

2 30-day backfill through the three layers

```
==> EPSS range 2026-09-21 .. 2026-10-20: 30 day(s)
  OK 30 file(s) passed the header, date and
     integrity gates
  OK trusted: every day landed with its source
     row count, 1 model version and 0 duplicate keys
  OK refined: 30 day(s), one row per vendor per day
```

3 query with measured cost, on all three layers

```
  raw      82325456 bytes scanned  -> USD 0.00037437
  trusted  56913236 bytes scanned  -> USD 0.00025881
  refined  162943 bytes scanned  -> USD 0.00004768
  answers identical across layers: true
  OK cost target met: 162943 bytes scanned
     <= 10485760 (USD 0.00004768)
```

4 idempotency and verifica.sh

```
  trusted.epss_scores
    rows 391915 -> 391915  (append-only would give 783830)
  refined.vendor_exposure_daily
    rows 290 -> 290  (append-only would give 580)

Resultado: 37 PASSA / 2 FALHA  (total: 39 criterios)
```

Every decision, one number

DECISION	THE NUMBER BEHIND IT
Grain (cve_id, day)	**0 duplicates in 11.5 M**
Daily partition	**29× fewer bytes (1 day)**
Parquet + ZSTD	**1.45× less read**
Pre-aggregated refined	**162,943 B · 64× under target**
Partition overwrite	**391,915 → 391,915**
Lake Formation per layer	**9 of 9 combinations**

FROM PROJECT TO PRODUCT

Prioritize patching by **real risk**: already exploited (KEV) and likely to be exploited now (EPSS), refreshed daily for **US$ 0.0003**.

ENGINEERING THAT SCALES

- SQL in Athena: **40×** cheaper than a Glue job.
- Daily incremental loads that never duplicate.

Repository: github.com/brunoribeirol/eda262-projeto-g07
All decisions: DECISIONS.md